Daikibo Industries

Operational & People Risk Report

Deloitte Data Analytics Job Simulation

May 2021 | Analyst: Brinda Jat

Tools Used: PostgreSQL | Microsoft Excel | Power BI

Client: Daikibo Industries — Global Manufacturing

Period: May 2021

Scope: 4 Factory Locations across Japan, Germany and China

Telemetry Records: 160,704 IoT machine messages

Equality Records: 37 roles across 4 factories

Tools: PostgreSQL, Microsoft Excel, Power BI

Analysis Type: Operational Risk + People Risk

Business Questions Answered

1. Which factory location had the most machine breakdowns in May 2021?
2. Which machine type broke down most often in that location?
3. Which factories and job roles show the most compensation discrimination?

Key Findings

Operational Findings

1. Daikibo Factory Seiko (Osaka, Japan) had the highest number of machine breakdowns — 48 out of 103 total breakdowns across all factories.

2. LaserWelder was the only machine type that broke down in Seiko factory — all 48 breakdowns came from this single machine type.

3. 89% of all breakdowns happened in the first 5 days of May 2021. After Day 7, breakdowns dropped to almost zero across all factories.

4. Daikibo Berlin had only 2 breakdowns the entire month — making it the most reliable factory and a good benchmark for others.

5. Machine temperature showed no link to breakdowns — healthy and unhealthy machines ran at almost the same temperature (25°C).

People Risk Findings

1. Daikibo Factory Meiyo has the worst average equality score of -14.36 — the most discriminative factory overall.

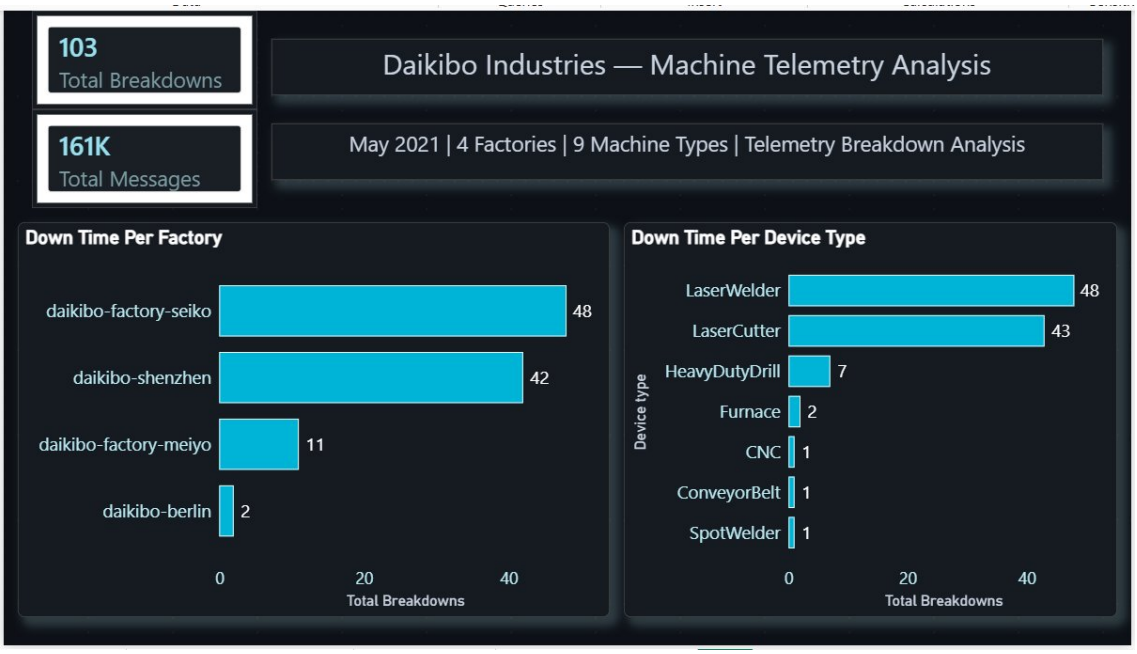
2. Every leadership role scores negatively. C-Level has the worst score at -25.0, followed by VP at -22.5.

3. Every technical and engineering role scores close to zero or positive — Jr. Engineer scored +3.5.

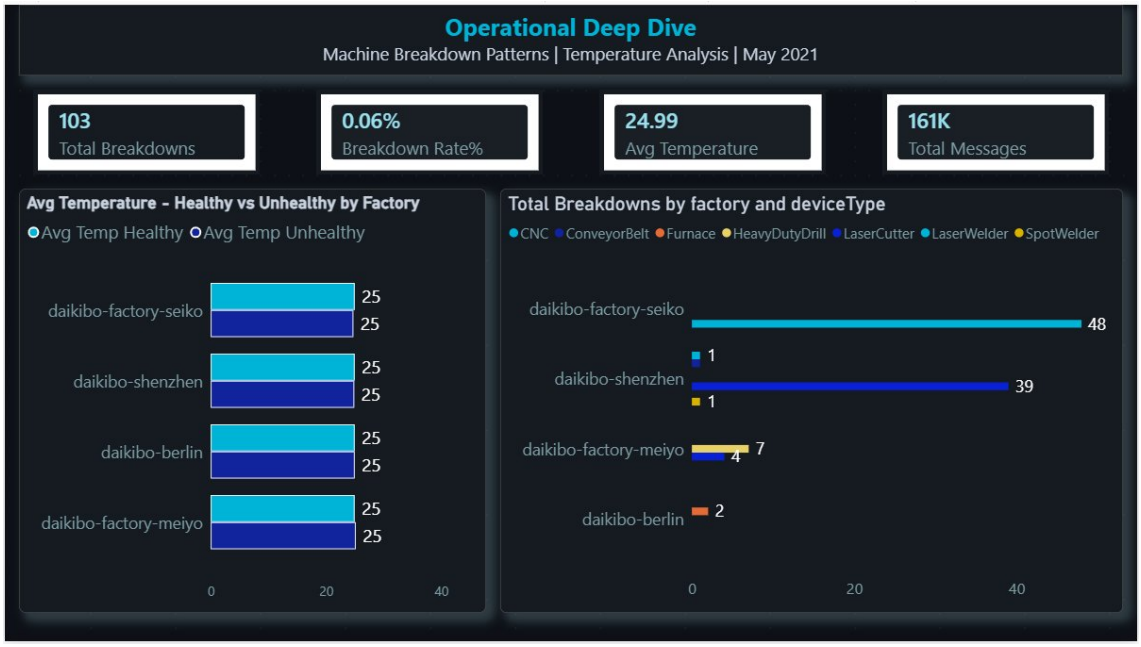
4. The higher the seniority, the worse the discrimination. This is a clear pattern across all 4 factories.

5. Daikibo Berlin has the best equality score (-5.50) and the best machine performance — suggesting management quality affects both.

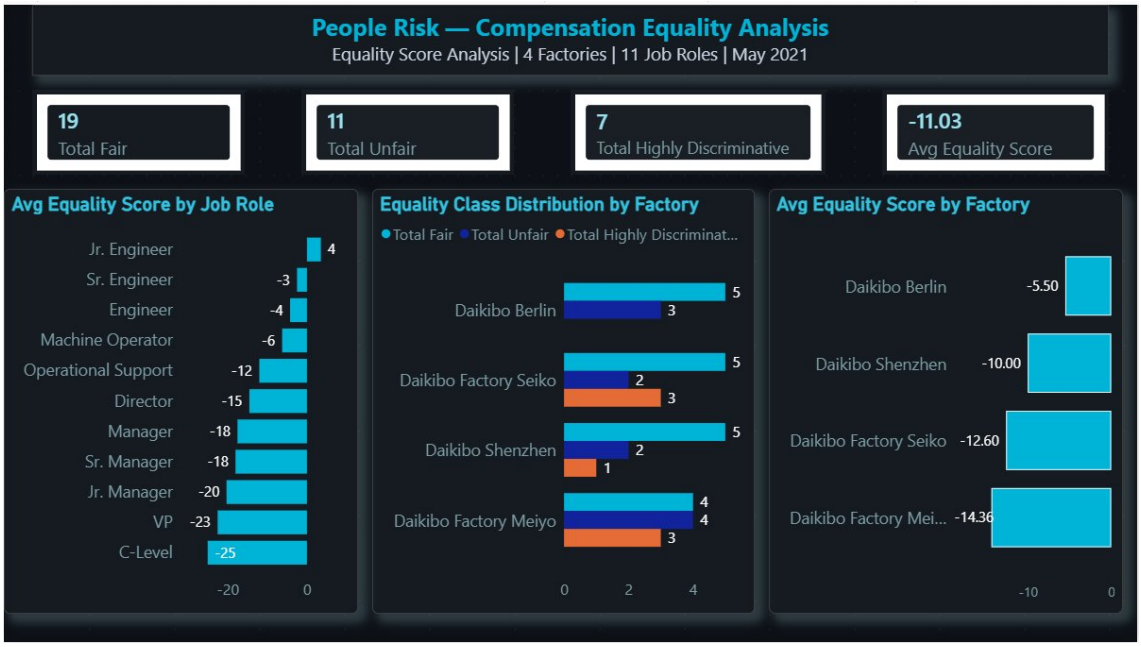
Power BI Dashboard



Page 1 — Factory Analysis: Down time per factory and per device type (Deloitte Simulation Output)



Page 2 — Operational Deep Dive: Breakdown patterns, temperature analysis and factory-machine combinations



Page 3 — People Risk: Equality score analysis by factory and job role

Page 4 — Executive Summary: Key findings and recommendations for the client

Recommendations & Risk Assessment

Recommendations

IMMEDIATE — Inspect and audit all LaserWelder machines at Daikibo Factory Seiko. One machine type caused 47% of all breakdowns across the entire company.

IMMEDIATE — HR and Legal team to review C-Level and VP compensation across all 4 factories. Both roles scored Highly Discriminative.

SHORT TERM — Study how Daikibo Berlin runs its operations and apply the same approach in Seiko and Meiyo factories.

SHORT TERM — Set up breakdown alerts during high-risk hours — especially 19:00 (shift change) and 03:00 (overnight with minimal staff).

LONG TERM — Run a monthly review of both telemetry data and equality scores to track if things are improving over time.

Risk Summary

High Risk — LaserWelder single point of failure at Seiko factory

High Risk — C-Level and VP compensation discrimination confirmed across all factories

High Risk — 48% of all job roles classified as Unfair or Highly Discriminative

Medium Risk — Shenzhen LaserCutter — 39 breakdowns, second highest after Seiko

Medium Risk — Breakdown spikes at shift change hours suggest supervision gaps

Low Risk — Berlin factory — near perfect on both machine performance and equality

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Data Limitations

- Only one month of data available — year on year trends cannot be confirmed yet.
- No cost data — financial impact of breakdowns could not be calculated.
- No demographic data in equality dataset — gender or age analysis was not possible.